

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 1 – Analytical Problem Solving (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Analytical problem solving
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
Student Name:	CH. Durga Prasad	Reg. No.:	192425305
Section / Batch:	B.Tech AIML 2024	Date:	16-07-2026

Code of Conduct

I **CH. Durga Prasad (192425305)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

CH. Durga Prasad

QUESTIONS: 5

Total Marks: 50

Annexure A

Analytical ProblemSolving

Duration: 1hr

Q1.

Describe the architecture of an Artificial Neural Network (ANN) consisting of input, hidden, and output layers. **Illustrate** the role of artificial neurons, weights, bias, and activation functions in processing input data to generate predictions. (10 marks).

An **Artificial Neural Network (ANN)** is a machine learning model inspired by the human brain. It consists of three main layers: the **input layer, hidden layer(s), and output layer**. The input layer receives the input features, such as age, income, or height, and passes them to the hidden layer. The hidden layer performs computations on the inputs using artificial neurons. An ANN may contain one or more hidden layers depending on the complexity of the problem. Finally, the output layer produces the prediction or classification result.

Each **artificial neuron** receives input values and multiplies them by their corresponding **weights**, which represent the importance of each input feature. A **bias** is then added to shift the output and improve the model's flexibility. The neuron computes the weighted sum using the equation:

$$z = \sum(W_i X_i) + b$$

The result is then passed through an **activation function** such as ReLU, Sigmoid, or Tanh. The activation function introduces non-linearity, allowing the network to learn complex relationships between inputs and outputs. Without an activation function, the network would behave like a simple linear model.

During training, the ANN adjusts its weights and biases using optimization algorithms such as Gradient Descent to minimize the prediction error. This enables the network to make accurate predictions for unseen data. ANNs are widely used in image recognition, speech processing, medical diagnosis, and financial forecasting because they can learn complex patterns from large datasets.

Q3.

Analyze the working of the Perceptron learning algorithm for solving a binary classification problem. **Demonstrate** the weight and bias update process for one iteration using a suitable numerical example. (10 marks)

The **Perceptron** is the simplest type of artificial neural network used for **binary classification** problems. It classifies data into one of two classes, such as Yes/No or Positive/Negative. The perceptron calculates a weighted sum of the inputs, adds a bias, and applies an activation function to generate the output.

The prediction is calculated as:

$$y = f(WX + b)$$

where W is the weight, X is the input, b is the bias, and f is the activation function.

Example:

Given:

- Input $x = 2$
- Target output $t = 1$
- Initial weight $W = 0.5$
- Bias $b = 0$
- Learning rate $\eta = 0.1$

First, compute the output:

$$z = (0.5 \times 2) + 0 = 1$$

Since $z > 0$, the predicted output is **1**.

Suppose the prediction was incorrect (predicted 0 instead of 1). The error is:

$$Error = Target - Output = 1 - 0 = 1$$

Update the weight:

$$\begin{aligned} W_{new} &= W + \eta \times Error \times x \\ W_{new} &= 0.5 + (0.1 \times 1 \times 2) = 0.7 \end{aligned}$$

Update the bias:

$$b_{new} = b + \eta \times Error = 0 + 0.1 = 0.1$$

Thus, the perceptron learns by updating its weights and bias whenever it makes an incorrect prediction. This process continues until the classification error is minimized.

Q4.

Illustrate the forward propagation process in a Multilayer Perceptron (MLP) using a simple neural network with one hidden layer. **Compute** the output values for the given inputs, weights, biases, and activation function. (10 marks)

Forward propagation is the process of passing input data through the layers of a neural network to obtain the final prediction.

Consider a simple MLP with:

- Input $x = 2$
- Weight $W_1 = 3$
- Bias $b_1 = 1$
- Weight $W_2 = 2$
- Bias $b_2 = 1$
- Activation Function = Identity

Step 1: Hidden Layer

Calculate the hidden neuron output:

$$\begin{aligned}z_1 &= W_1x + b_1 \\z_1 &= (3 \times 2) + 1 = 7\end{aligned}$$

Since the identity activation function is used:

$$a_1 = 7$$

Step 2: Output Layer

Calculate the final output:

$$\begin{aligned}z_2 &= W_2a_1 + b_2 \\z_2 &= (2 \times 7) + 1 = 15\end{aligned}$$

Using the identity activation:

$$\hat{y} = 15$$

Therefore, the network predicts an output of **15**.

Forward propagation allows information to move from the input layer through the hidden layer to the output layer. The prediction is then compared with the actual value to calculate the loss during training.

Q5.

Demonstrate the Backpropagation algorithm for training a neural network by calculating the output error, propagating the error through hidden layers, and updating the network weights using Gradient Descent. (10 marks)

Backpropagation is a supervised learning algorithm used to train neural networks. It minimizes the prediction error by updating the weights and biases using the **Gradient Descent** optimization algorithm.

Consider:

- Input $x = 4$
- Target output $y = 15$
- Weight $W = 2$
- Bias $b = 1$
- Learning rate $\eta = 0.1$

First, calculate the prediction:

$$\hat{y} = Wx + b = (2 \times 4) + 1 = 9$$

The loss using Mean Squared Error is:

$$L = \frac{1}{2}(15 - 9)^2 = 18$$

The gradient of the loss with respect to the weight is:

$$\begin{aligned}\frac{\partial L}{\partial W} &= (\hat{y} - y)x \\ &= (9 - 15) \times 4 = -24\end{aligned}$$

Update the weight:

$$\begin{aligned}W_{new} &= W - \eta \frac{\partial L}{\partial W} \\ &= 2 - 0.1(-24) = 4.4\end{aligned}$$

Similarly, update the bias:

$$\begin{aligned}\frac{\partial L}{\partial b} &= \hat{y} - y = -6 \\ b_{new} &= 1 - 0.1(-6) = 1.6\end{aligned}$$

The updated weights and bias reduce the prediction error. This process is repeated for multiple iterations until the network reaches minimum loss and produces accurate predictions.

Q6.

Compare and evaluate Batch Gradient Descent, Stochastic Gradient Descent (SGD), and Mini-Batch Gradient Descent with respect to convergence behavior, computational efficiency, memory requirements, and practical applications in deep learning. (10 marks)

Gradient Descent is an optimization algorithm used to minimize the loss function by updating the weights of a neural network. The three commonly used variants are **Batch Gradient Descent**, **Stochastic Gradient Descent (SGD)**, and **Mini-Batch Gradient Descent**.

Batch Gradient Descent uses the entire training dataset to calculate the gradient before updating the weights. It provides stable and accurate convergence but is computationally expensive and requires high memory. It is mainly used for small datasets.

Stochastic Gradient Descent (SGD) updates the weights after processing each training sample. It requires very little memory and is much faster than Batch Gradient Descent. However, its updates are noisy, causing fluctuations during convergence. SGD is suitable for very large datasets and online learning.

Mini-Batch Gradient Descent combines the advantages of both methods by dividing the dataset into small batches (such as 32 or 64 samples). The weights are updated after each batch, resulting in faster training, lower memory usage, and smoother convergence than SGD. It is the most widely used optimization method in modern deep learning applications.

Feature	Batch Gradient Descent	Stochastic Gradient Descent (SGD)	Mini-Batch Gradient Descent
Data Used	Entire dataset	One sample	Small batch
Convergence	Stable but slow	Fast but noisy	Fast and stable
Memory Requirement	High	Low	Moderate
Computational Efficiency	Low	High	High
Applications	Small datasets	Online learning	Deep learning, CNNs, RNNs

Overall, **Mini-Batch Gradient Descent** is preferred in deep learning because it provides a good balance between convergence speed, computational efficiency, and memory usage while achieving high model performance.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, wellanalyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____